

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

- <think>
- First , use the formula $KE = p^2 / (2m)$ to find the momentum of the electron .
 - Then , use de Broglie's formula , $\lambda = h / p$, to find the wavelength .
 - 1 eV corresponds to 1.602×10^{-12} erg .
 - Use the electron mass and conversion factor to find wavelength in cm .
 - Checking the closest match with the provided answer .

</think>

<answer>

A

</answer>

